

Harsh Agarwal

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PROFILE

Computer Science graduate with hands-on experience in Machine Learning, Deep Learning, and Generative AI. Experienced in building end-to-end AI applications using Python, LangChain, LangGraph, and RAG.

EDUCATION

Bachelor of Technology in Computer Science and Engineering

Arya College of Engineering and IT, Jaipur | 2022 – 2026

CGPA - **8.31**

Senior Secondary (Class XII), CBSE

MPS International School, Jaipur | 2021 – 2022

Percentage - **80.5%**

TECHNICAL SKILLS

Languages: Python, C++

Database: MySQL

Libraries & Frameworks: NumPy, Pandas, Seaborn, Scikit-learn, TensorFlow

Generative AI: LangChain, LangGraph, RAG, Vector Databases

Machine Learning: Supervised & Unsupervised Learning, Feature Engineering

Deep Learning: ANN, CNN, RNN, LSTM, GRU

PROJECTS

Agentic RAG Chatbot — Multi-Tool AI Assistant

Tech Stack: Python, LangGraph, LangChain, Groq (Llama 3.3 70B), FAISS, Hugging Face Embeddings, Streamlit
<https://github.com/harshaga819/agentic-rag-chatbot.git>

- Built a conversational AI assistant that handles normal chat, document-based Q&A, and live tool use within the same conversation, automatically switching based on what the user asks.
- Enabled users to chat with an uploaded PDF or any website link, using vector search to answer questions based on the document's or page's actual content.
- Integrated real-time tools, including web search and live stock price lookup, so the assistant can answer questions beyond static documents when needed.
- Added persistent, multi-thread chat memory so users can maintain and resume multiple separate conversations, each with its own uploaded content, across sessions.

Facial Emotion Recognition System — Deep Learning-Based Emotion Classification

Tech Stack: TensorFlow, Keras, OpenCV, MobileNetV2, transfer learning
<https://github.com/harshaga819/Facial-Emotion-Recognition-System.git>

- Developed a deep learning-based facial emotion recognition system leveraging **transfer learning** with MobileNetV2 architecture.
- **Preprocessed** and **augmented** the CK+ dataset to improve model generalization and **reduce overfitting**.
- Built a multi-class classification pipeline to detect 7 distinct emotion categories from **facial images**.
- Achieved approx **80%** validation accuracy, demonstrating effective **feature extraction** and **model tuning**.

CERTIFICATES

SQL (Intermediate) - Hacker-Rank

Introduction to Generative AI - SkillUP

ACHIEVEMENTS

Solved **250+** DSA problems on LeetCode and HackerRank

Earned **4 Star** in C++ and **3 Star** in SQL on HackerRank